Surveillance system

The security camera by Prama Hikvision incorporates AcuSense technology that precisely captures any unwanted intrusion. It reduces false alarm triggering, does more efficient and effective file searching (based on human and vehicle classification), and combines siren with flashing light. With high accuracy, the system disregards alarms triggered by other objects such as rain or leaves and delivers alarms that are associated with human or vehicle detection.

Prama Hikvision
www.hikvision.com

Component epoxy

The Vitralit UD 1405 from Panacol-Elosol GmbH is a transparent, solvent-free, one-component epoxy that has low viscosity and good wetting properties. It allows the reinforced windings to be cured within seconds through irradiation with UV (365nm) or visible (405nm) light. Fibres or filaments can first be pulled through an adhesive bath, then wound onto the rotor or respective cylinder and cured with UV or visible light. A secondary thermal cure ensures that the layers of coated windings below the surface become fully cured. After curing, Vitralit UD 1405 is electrically insulating, shows minimal shrinkage, and remains resistant at temperatures from -40 degrees to 180 degrees Celsius.

Panacol-Elosol GmbH
www.panacol.com

MANUFACTURING EQUIPMENT

3D printer

The Ultimaker 2+ Connect 3D printer from Ultimaker allows users to send print jobs via Wi-Fi or Ethernet. Cloud 3D printing enables remote file transfer with added security from anywhere globally. The printer also offers a 6cm colour touchscreen, an ergonomic feeder lever, and a stiffer build platform for ease of use and reliability.

Ultimaker
www.ultimaker.com

Microwave amplifier

The R&S SAM100 microwave amplifier from Rohde & Schwarz features ease of operation, robust design, and super-compact footprint within its 2 to 20GHz range with up to 20W output power. It targets manufacturers of passive and active microwave components, and microwave devices for mobile radio (UMTS, LTE, 4G, 5G), IoT (WLAN, Bluetooth), satellite, and radar applications. Its high gain with low noise at superior linearity makes the unit highly suited to AM, FM, PM, and OFDM applications.

Rohde & Schwarz
www.rohde-schwarz.com

MODULES & BOARDS

IoT modules

Silicon Labs has launched wireless modules with full-stack support for commercial and consumer IoT application development with flexible package options

and highly integrated device security. The new modules include xGM210PB that supports Bluetooth, Zigbee, OpenThread, and Wi-Fi; BGM220—ideal for a broad range of Bluetooth 5.2 LE applications including appliances, asset tags, beacons, portable medical, fitness and Bluetooth mesh low-power nodes; MGM220 - ideal for Zigbee Green Power and energy harvesting applications; BGX220 Xpress that includes an onboard Bluetooth stack, Xpress command interface, and pre-programmed cable replacement firmware to deliver a serial-to-Bluetooth LE solution needing no firmware development.

Silicon Labs
www.silabs.com

Haptic control board

There is a growing demand for vibrational feedback, which allows more intuitive control. The HAPTIC Reactor Heavy Type developed by Alps Alpine produces an excitation force of 15G (at 100g) at a low frequency (130Hz), despite compact dimensions of 33 × 23 × 13mm (W × D × H), while retaining the typical HAPTIC Reactor characteristics of horizontal resonance, which is easier to perceive with fingertips and has an outstanding response speed. A small input voltage of 7V eliminates the need for a booster circuit.

Alps Alpine
www.alpsalpine.com

SoM

PolarBerry, a production- and deployment-ready SoM by Sundance Multiprocessor comes with a hardened 64-bit, multicore real-time, Linux-capable RISC-V MPU subsystem and PolarFire SoC field-programmable gate array (FPGA). Its compute engine delivers up to 50% lower power than alternative FPGAs, 250k logic elements (LEs), and features four high-speed, low-power transceivers from 250Mbps to 12.7Gbps. It is suited to the development of embedded computing and real-time applications based on RISC-V processors as an alternative to ARM. Typical applications will be found in the defence/mil-aero, AI/ML, communications, automotive, industrial automation, and imaging markets as well as the Internet of Things.

Sundance Microprocessor Technology
www.sundance.com

Single board computer

To be in line with the fast-growing semiconductor tech for various IoT applications, the open-source low-cost RuggedBOARD SBC with phyCORE-A5D2x SoM based on ARM Cortex-A5 SoC @500 MHz, provides a good plat-